A PbWO4 SCINTILLATOR RANGE-TELESCOPE FOR DEPTH-DOSE MEASUREMENTS IN LOW-FLUX ENVIRONMENTS

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Faculty Disclosure

X	No, nothing to disclose
	Yes, please specify:

Company Name	*Honoraria/ Expenses*	*Consulting/ Advisory Board*	*Funded Research*	*Royalties/ Patent*	*Stock Options*	*Ownership/ Equity Position*	*Employee*	*Other (please specify)*

Why Low-Dose Environments?

- Conventional dosimetry systems are designed for high-dose environments
  - Water columns (PTWPeakFinder)
  - MLIC (IBA Dosimetry Giraffe & Zebra)

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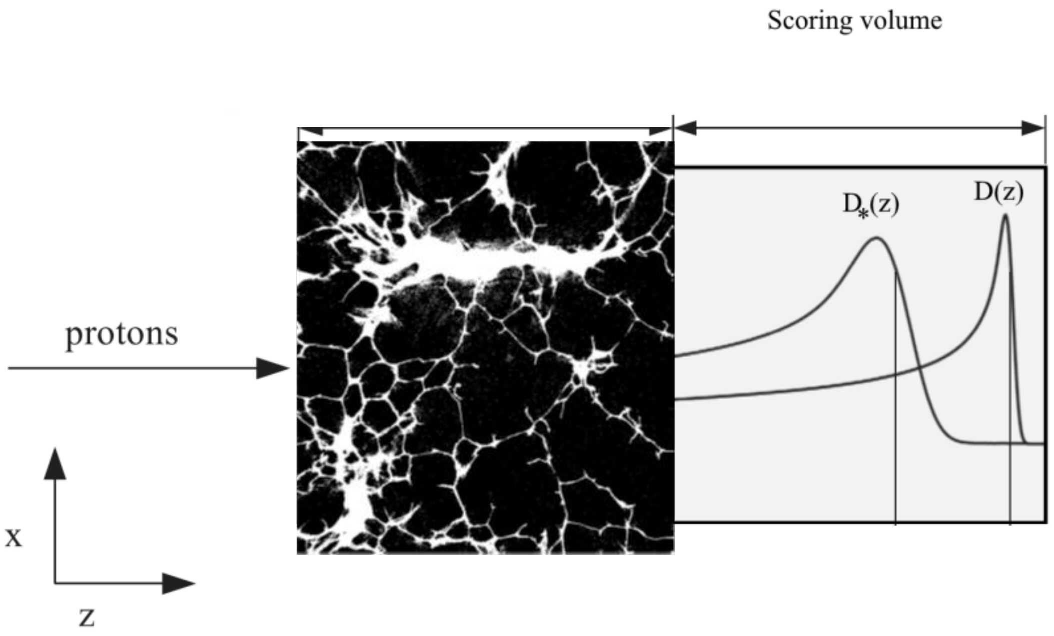
- Low-flux environments offer individual particle tracking
  - Useful for detector development, Monte Carlo validation
  - Direct low-dose measurements of heterogeneous phantoms



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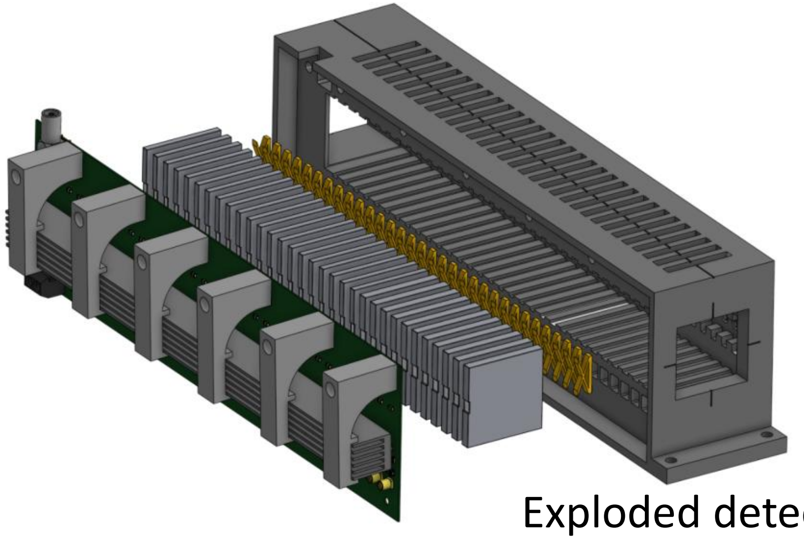
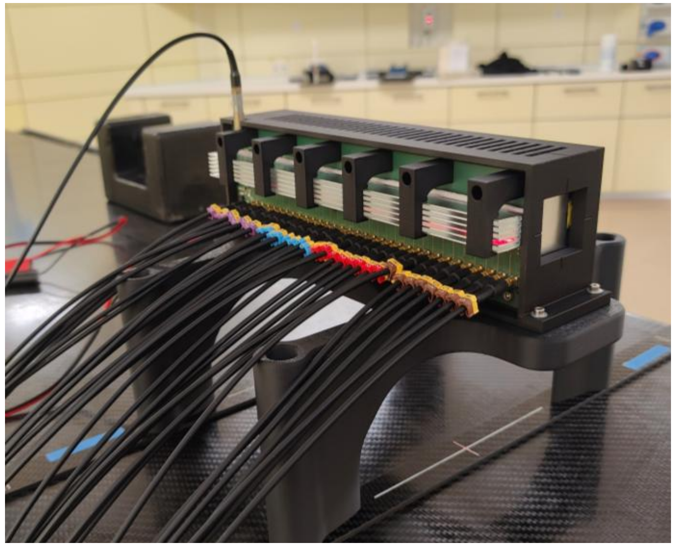
- Optical coupling with Baysilone Fluid M and pressure using 3D-printed springs

Wrapped $\mathrm{PbWO_4}$-Crystal

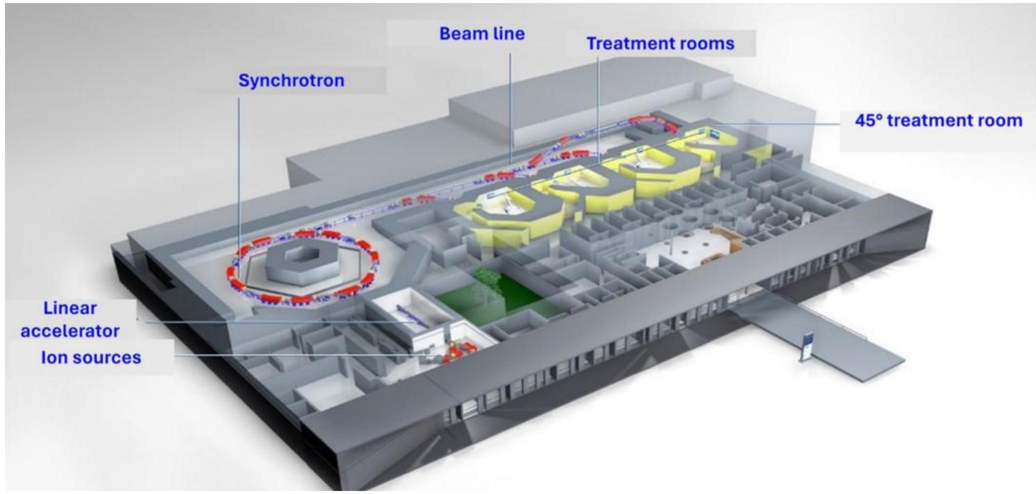




Exploded detector Schematic & Prototype

Beam Tests

- Measurements at Ion-Beam Therapy Center (MIT) in Marburg



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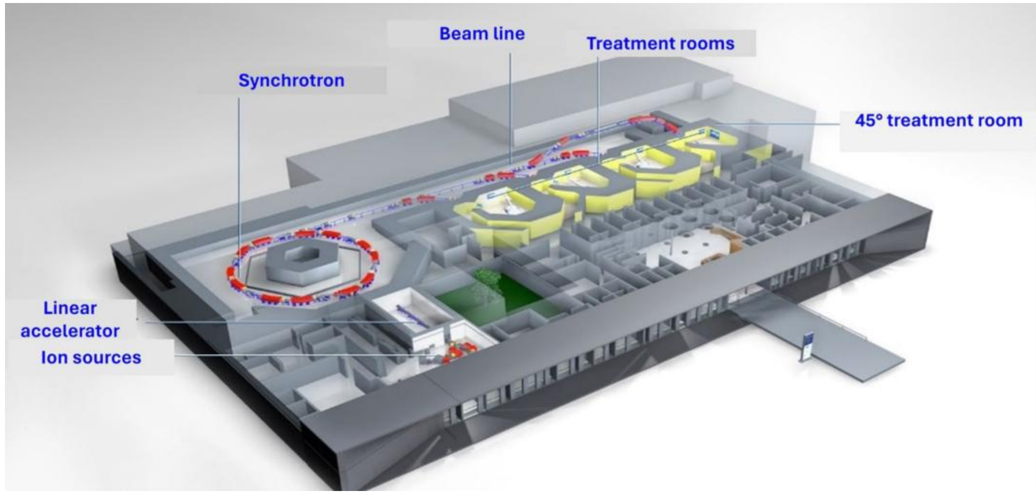
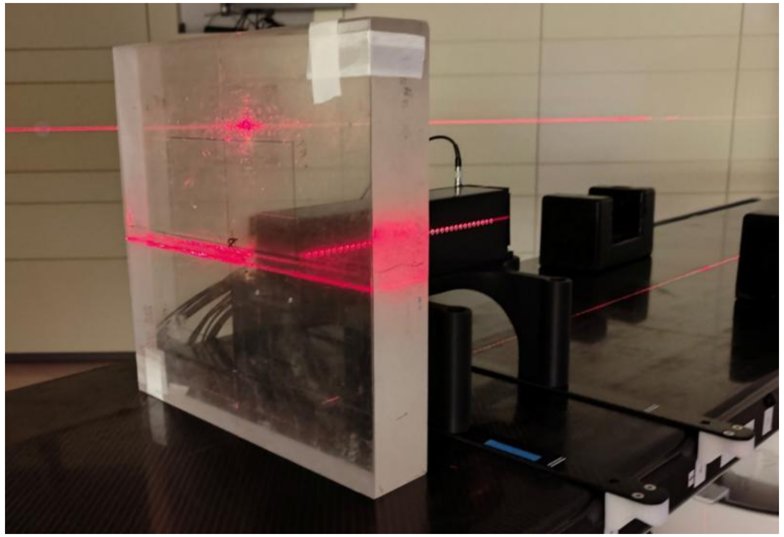
  - Homogeneous PMMA phantom 52 mm to 57 mm in 1 mm steps





PMMA Phantom

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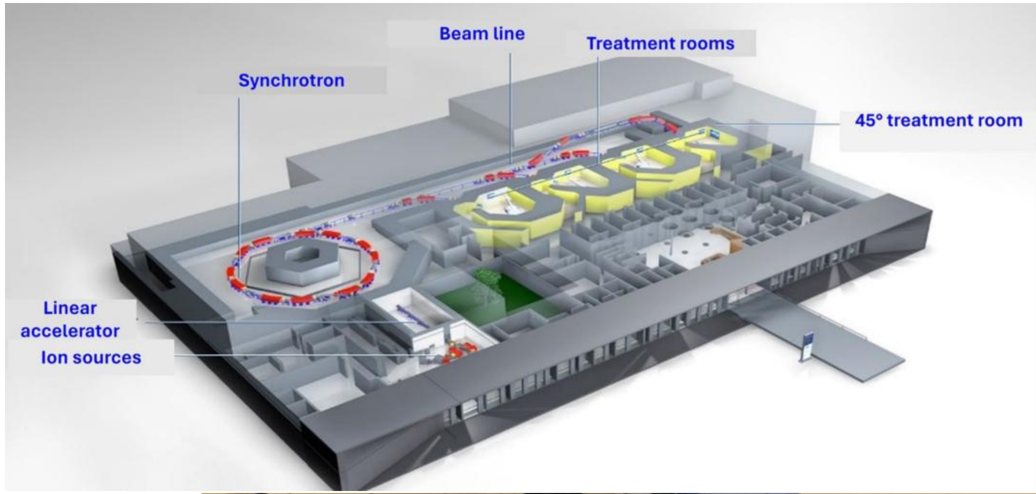
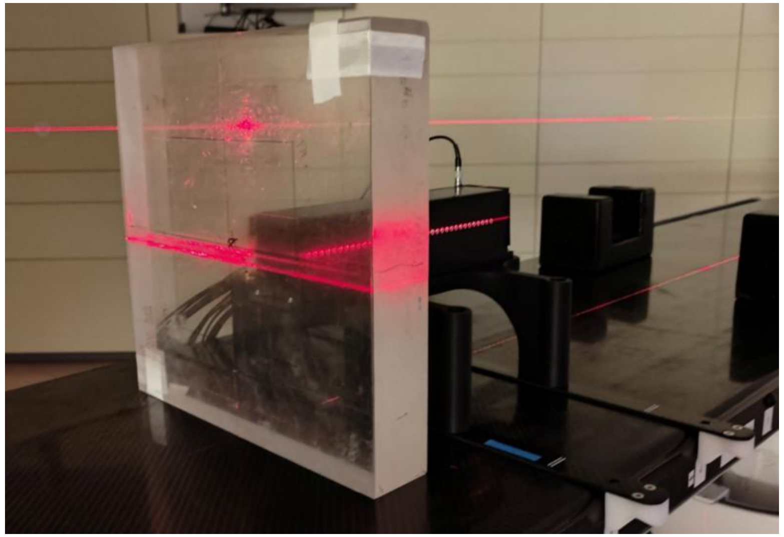
- Measurements at Ion-Beam Therapy Center (MIT) in Marburg
  - Homogeneous PMMA phantom 52 mm to 57 mm in 1 mm steps
  - Heterogeneous LN300 phantom 200 mm





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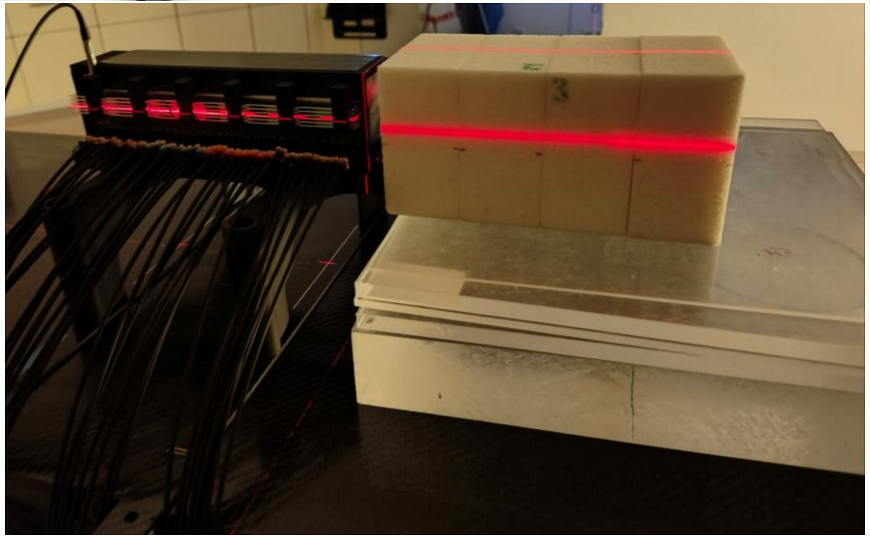
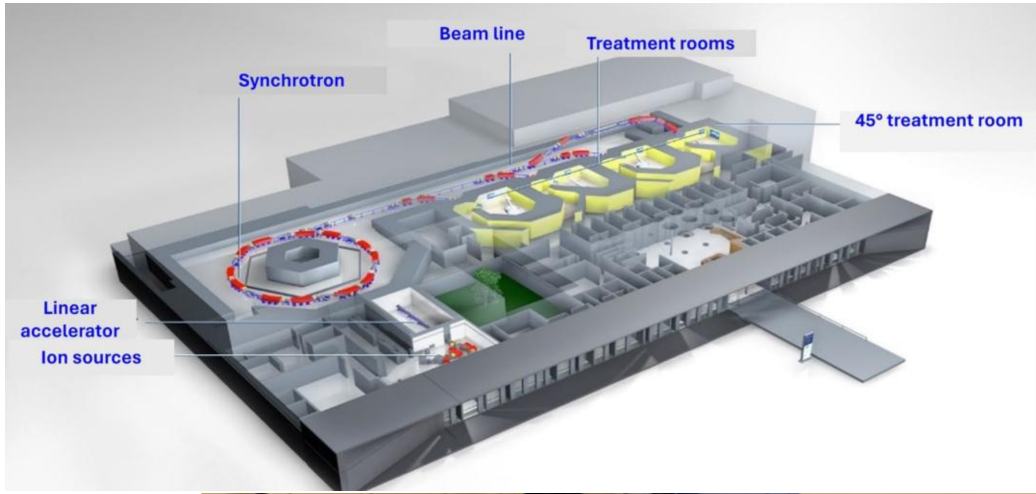
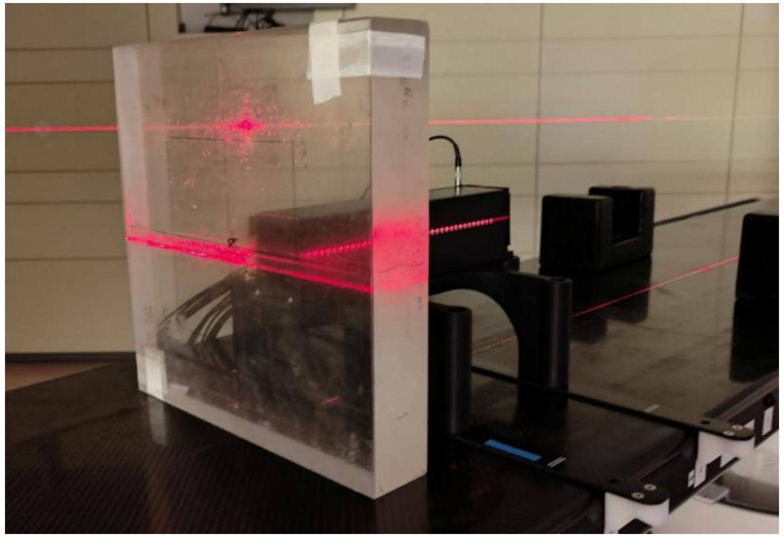


LN300 Phantom

Beam Tests

- Measurements at Ion-Beam Therapy Center (MIT) in Marburg
  - Homogeneous PMMA phantom $52\ \mathrm{mm}$ to $57\ \mathrm{mm}$ in $1\ \mathrm{mm}$ steps
  - Heterogeneous LN300 phantom $200\ \mathrm{mm}$
- Proton beam at $221.6\ \mathrm{MeV}$ with particle flux of $2300\,\frac{1}{\mathrm{s}}$





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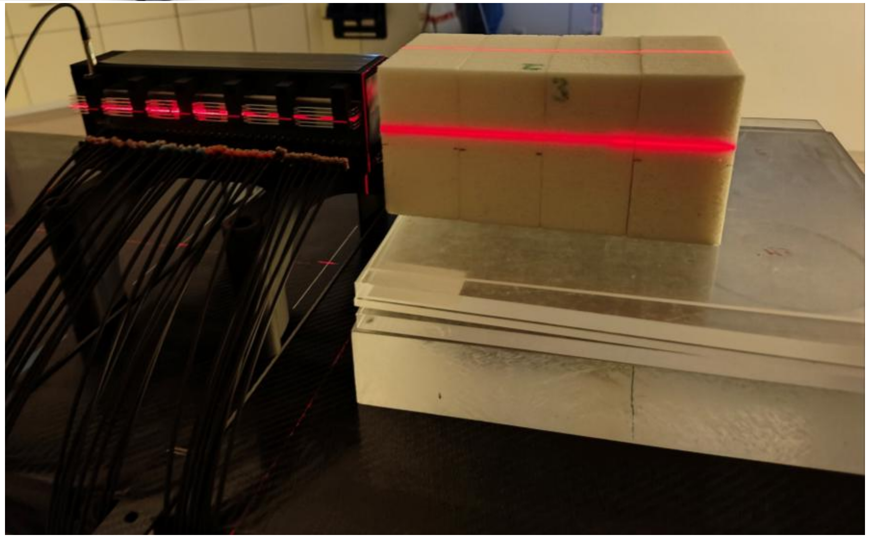

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- Protons classified as multiple energy deposition events within a $50\ \mathrm{ns}$ global coincidence window
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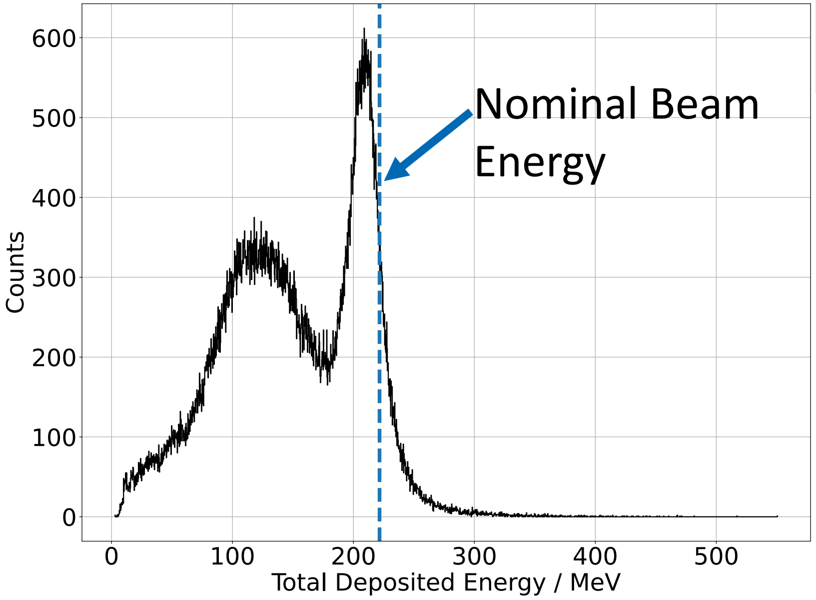

Data Analysis

- Protons classified as multiple energy deposition events within a 50 ns global coincidence window
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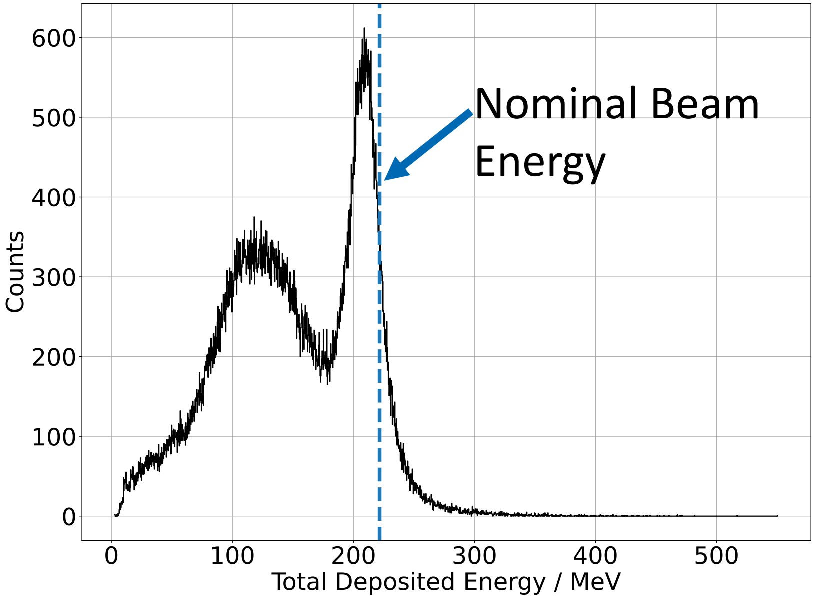
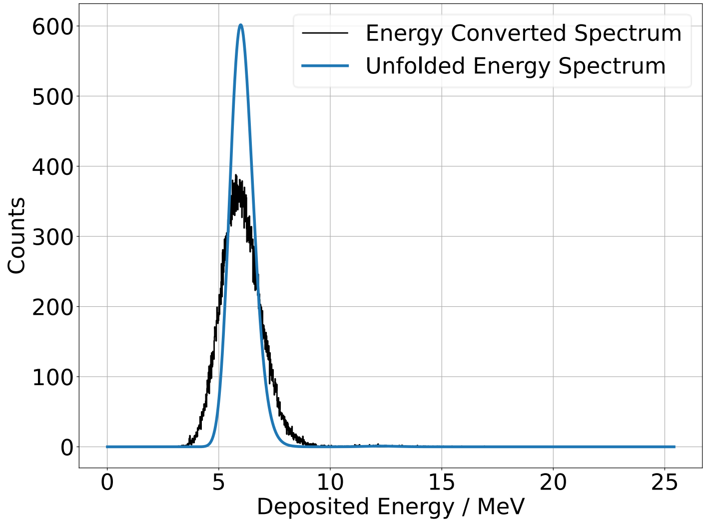
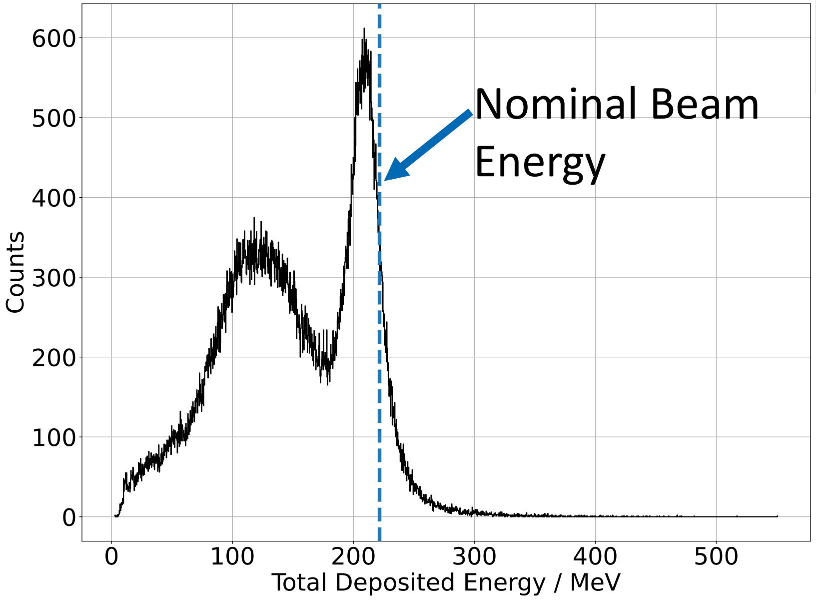
- Energy deposition per channel unfolded
  - Accounts for photon statistics and microcell saturation



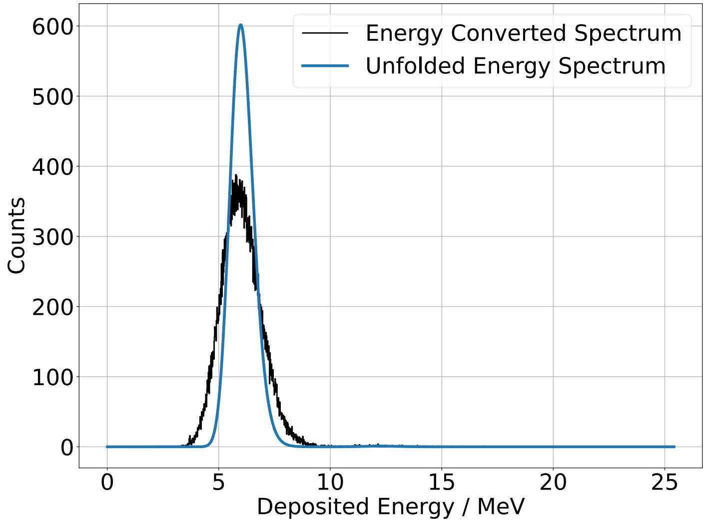


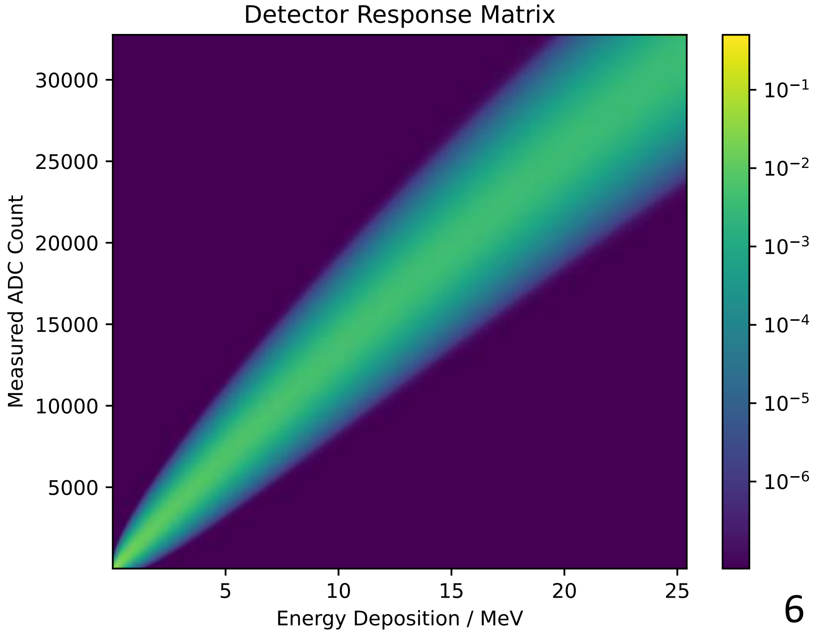
Data Analysis

- Protons classified as multiple energy deposition events within a 50 ns global coincidence window
  - Main peak of total energy deposition using in analysis
- Energy deposition per channel unfolded
  - Accounts for photon statistics and microcell saturation
- Detector response function used for Bayesian unfolding

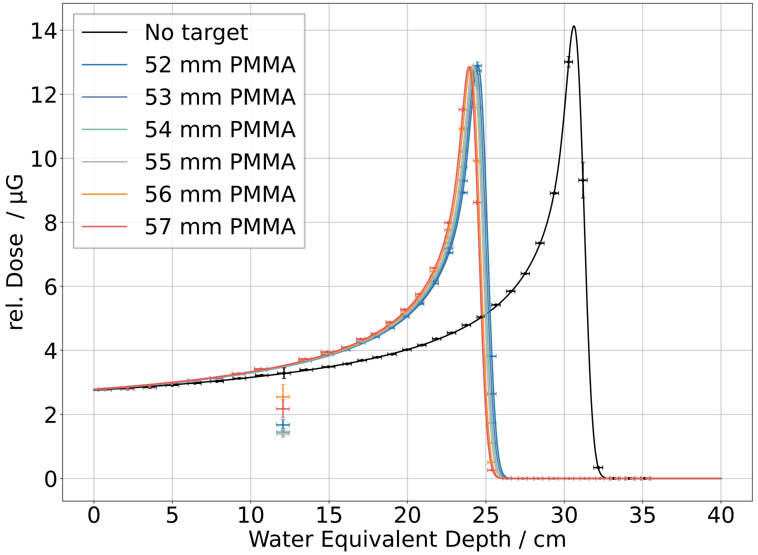


$$R_{ij} = P(A_i \mid E_j) \approx \sum_{k=0}^{k_{\max}} \binom{N_{\text{cells}}}{k} p(E_j)^k (1 - p(E_j))^{N_{\text{cells}} - k} \cdot \mathcal{N}(A_i - k \cdot \mu_{\text{SPE}},\ 0,\ \sqrt{k}\sigma_{\text{SPE}})$$





Results



- Fitted using analytical approximation of T. Bortfeld (1997)

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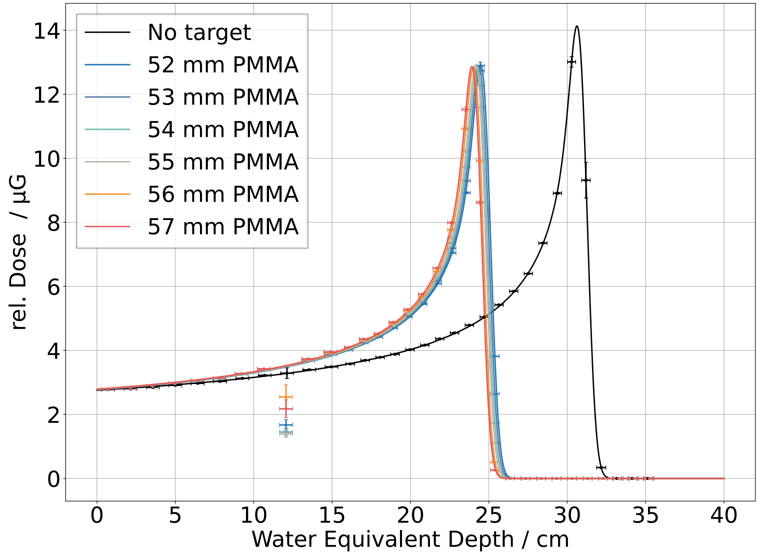

- Fitted using analytical approximation of T. Bortfeld (1997)
- WET Phantom thickness reconstruction within 0.3 mm (0.5 %)

Phantom	R_0 / cm	σ / cm	t / cm	$t_{\text{nom.}}$ / cm	Δt / µm	$\Delta t_{\%}$ / %
None	30.997 ± 0.002	0.453 ± 0.005	-	-	-	-
52 mm	24.850 ± 0.003	0.449 ± 0.003	6.147 ± 0.004	6.136	110	0.18
53 mm	24.717 ± 0.004	0.445 ± 0.002	6.280 ± 0.004	6.254	260	0.41
54 mm	24.601 ± 0.004	0.441 ± 0.003	6.396 ± 0.004	6.372	240	0.38
55 mm	24.493 ± 0.004	0.435 ± 0.003	6.504 ± 0.004	6.490	140	0.22
56 mm	24.362 ± 0.004	0.441 ± 0.006	6.635 ± 0.004	6.608	270	0.41
57 mm	24.253 ± 0.003	0.440 ± 0.008	6.744 ± 0.003	6.726	180	0.27

N. Fiedler et al., JINST 2026 DOI: 10.1088/1748-0221/21/05/P05018

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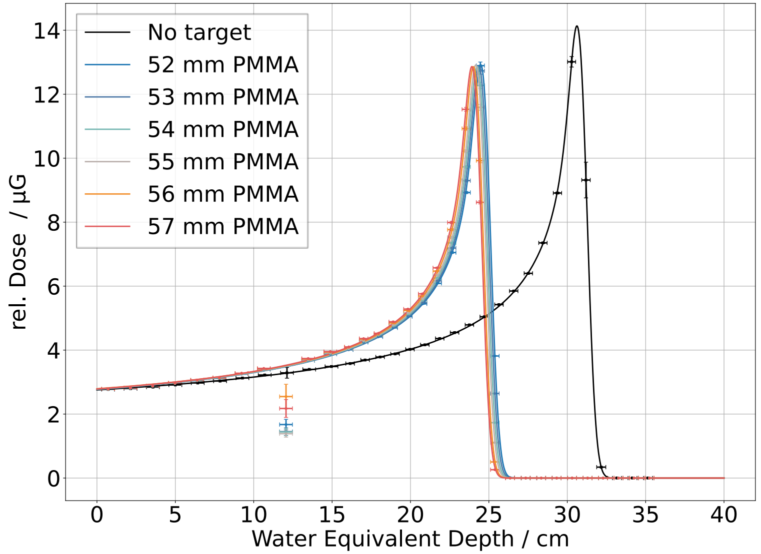

- Fitted using analytical approximation of T. Bortfeld (1997)
- WET Phantom thickness reconstruction within 0.3 mm (0.5 %)
  - ±0.4 mm to 0.5 mm for IBA Giraffe/Zebra/Lynx, Sun Nuclear DQA3

Phantom	R_0 / cm	σ / cm	t / cm	$t_{nom.}$ / cm	Δt / µm	$\Delta t_{\%}$ / %
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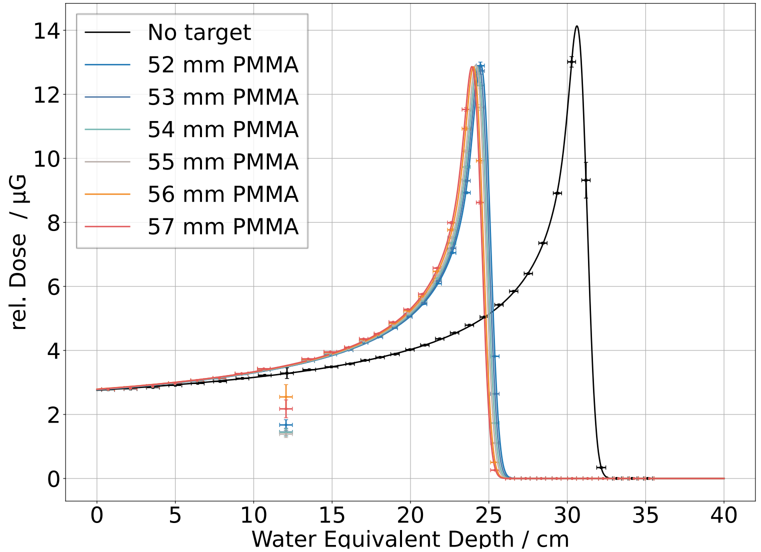

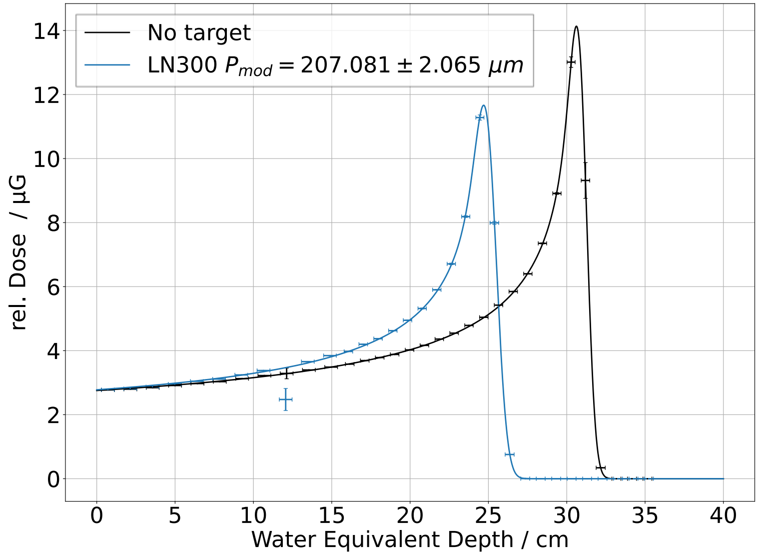

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  - $\pm$0.4 mm to 0.5 mm for IBA Giraffe/Zebra/Lynx, Sun Nuclear DQA3
- $P_{\text{mod, meas.}} = 207.1\ \mu\text{m}$, $P_{\text{mod, nom.}} = 210\ \mu\text{m}$

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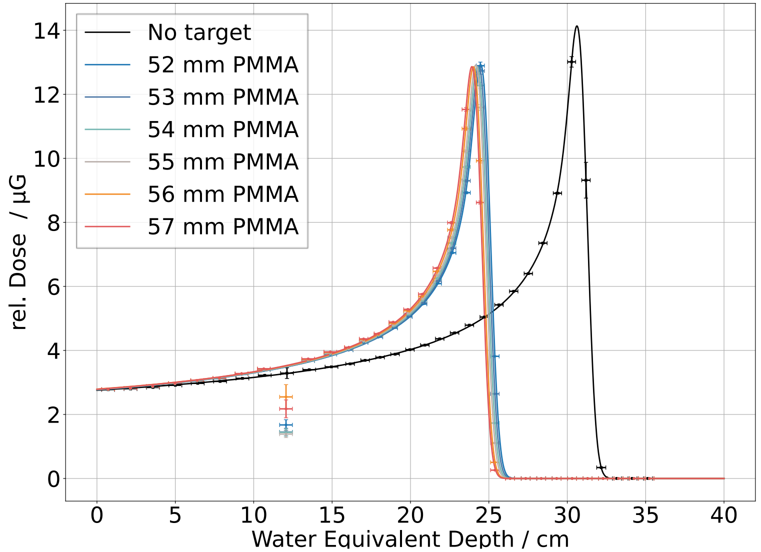

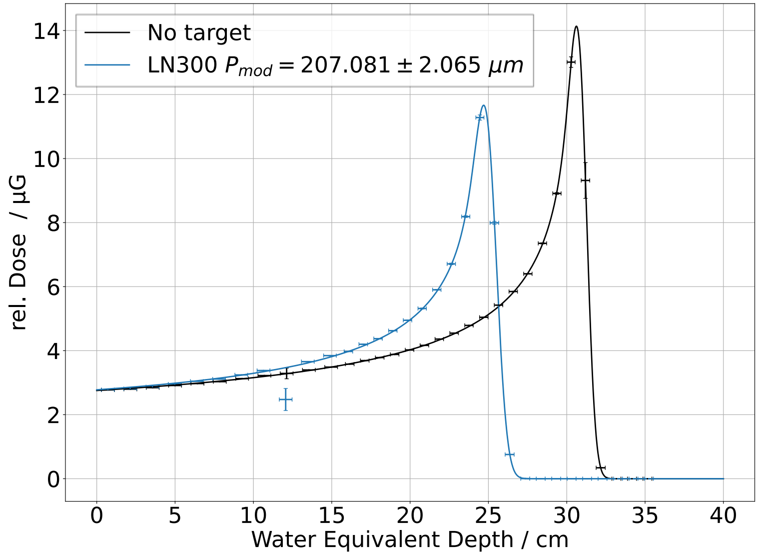

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- $P_{\mathrm{mod,\,meas.}} = 207.1\ \mu\mathrm{m}$, $P_{\mathrm{mod,\,nom.}} = 210\ \mu\mathrm{m}$
- Cumulative phantom energy deposition < 2 µJ
  - Reducible by at least a factor 10

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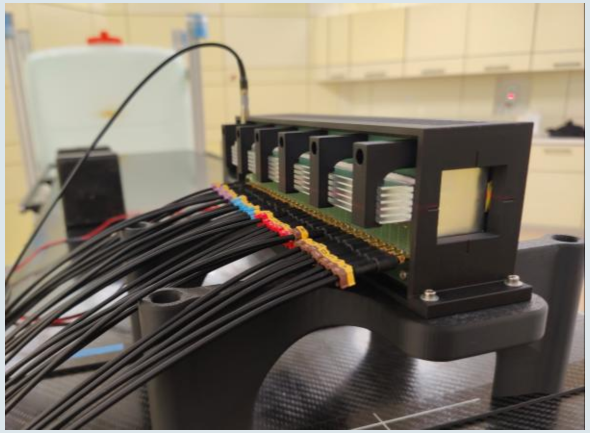

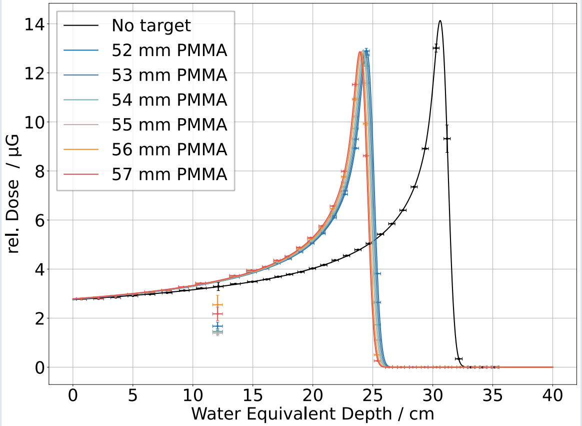

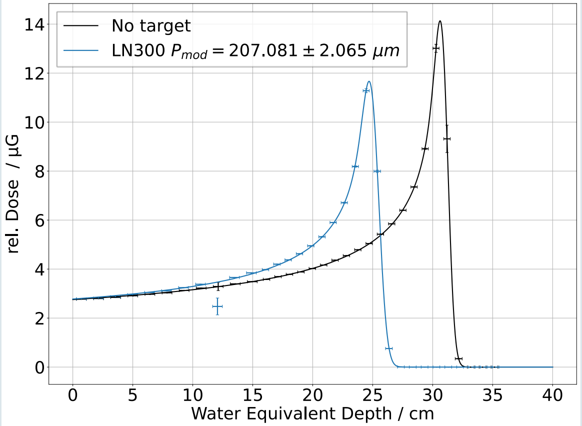

Thank you for your attention!